Segment-6
Operational Amplifier

Aneela Ahmed (a233491)

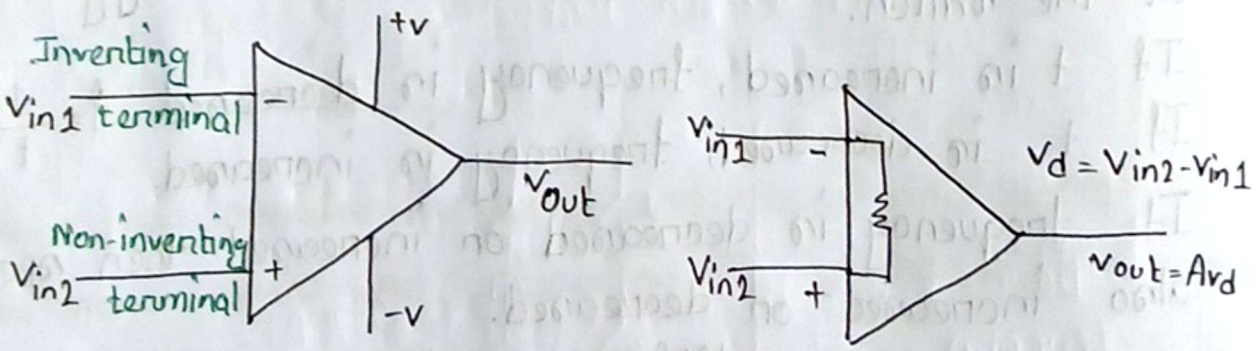
Op-amp:-

An operational amplifier is an integrated circuit that can amplify weak electric signals. It can perform some mathematical function like addition, subtraction, differenciation, integration etc.

Operational Amplifier:-

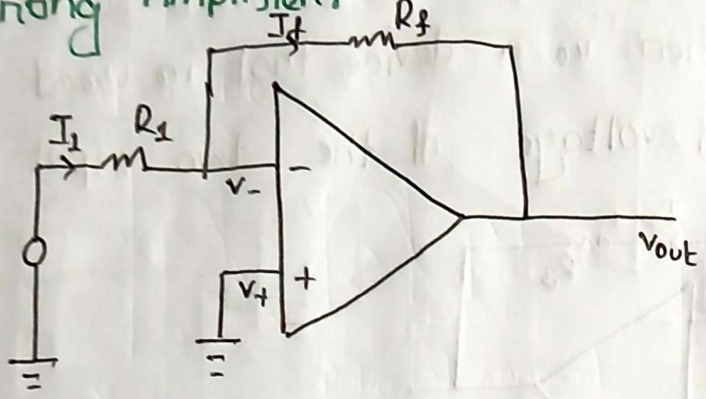
- → Additiona
- → Subtraction
- → Differentiation
- → Integration
- → voltage follower
- → Filter
- → Oscillator
- → voltage comparator.

An op-amp has high input impedence (∞) and low output impedence (0).



Operational amplifier has two inputs 1. Inverting
2. Non inverting

Inverting Amplifier:-



Applying KCL,

$$I_1 - I_f = 0$$

$$\Rightarrow I_1 = I_f$$

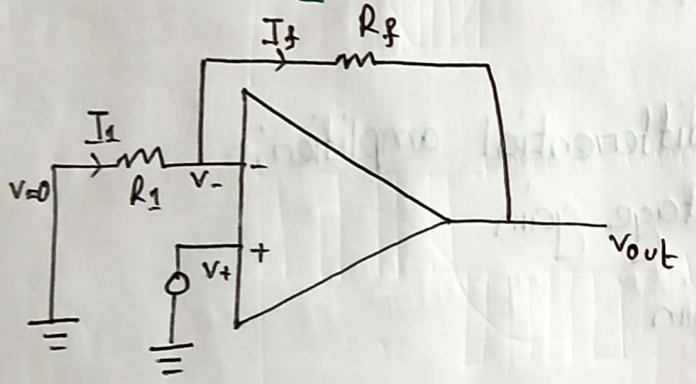
$$\Rightarrow \frac{V_{in} - V^-}{R_1} = \frac{V^- - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{in} - 0}{R_1} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{in}}{R_1} = \frac{-V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_1}$$

Non-inverting amplifier:-



Here,

$V_{in} = V^+$

$\therefore V^- = V_{in}$

$V^- - V^+ = 0$

$\Rightarrow V^- = V^+$

$$I_1 = I_f$$

$$\Rightarrow \frac{V - V_{in}}{R_1} = \frac{V_{in} - V_{out}}{R_f}$$

$$\Rightarrow \frac{0 - V_{in}}{R_1} = \frac{V_{in} - V_{out}}{R_f}$$

$$\Rightarrow -V_{in}R_f = V_{in}R_1 - V_{out}R_1$$

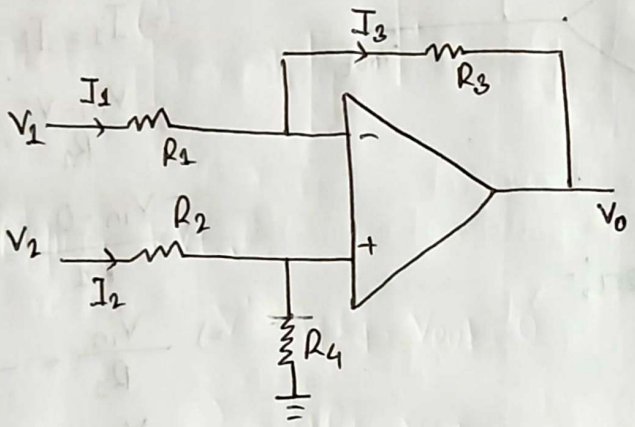
$$\Rightarrow V_{in}(R_f + R_1) = V_{out}R_1$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{R_1 + R_f}{R_1}$$

$$= 1 + \frac{R_f}{R_1}$$

Differential Amplifier:-

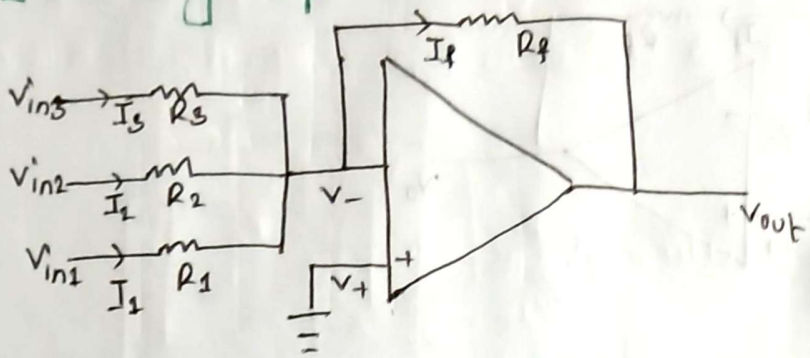
Differential amplifier is a device that is used to amplify the difference in voltage of the two input signals.



Characteristics of differential amplifier:-

1. High differential voltage gain
2. Low common mode gain
3. High input impedence
4. Low output impedence
5. High CMRR and large bandwidth
6. Low offset voltages and current.

Summing Amplifier:



$$V^- - V^+ = 0$$

$$\Rightarrow V^- = V^+$$

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_{in1}-0}{R_1} + \frac{V_{in2}-0}{R_2} + \frac{V_{in3}-0}{R_3} = \frac{0-V_{out}}{R_f}$$

$$\Rightarrow \frac{V_{in1}}{R_1} + \frac{V_{in2}}{R_2} + \frac{V_{in3}}{R_3} = \frac{-V_{out}}{R_f}$$

$$\Rightarrow V_{out} = -R_f\left(\frac{V_{in1}}{R_1} + \frac{V_{in2}}{R_2} + \frac{V_{in3}}{R_3}\right)$$

$$\Rightarrow V_{out} = -R\left(\frac{V_{in1}}{R} + \frac{V_{in2}}{R} + \frac{V_{in3}}{R}\right) \quad [R_f = R_1 = R_2 = R_3 = R]$$

$$\Rightarrow V_{out} = -(V_{in1} + V_{in2} + V_{in3})$$

$$\therefore \frac{V_{out}-V}{R} = \frac{V-V_{out}'}{R}$$

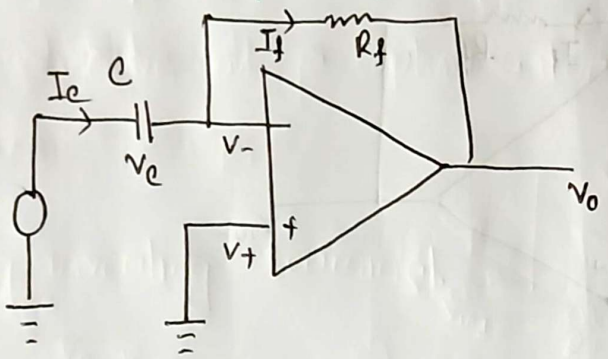
$$\Rightarrow V_{out} = -V_{out}'$$

$$\Rightarrow V_{out}' = -V_{out}$$

$$\Rightarrow V_{out}' = -[-(V_{in1} + V_{in2} + V_{in3})]$$

$$= V_{in1} + V_{in2} + V_{in3} \text{ (Ans)}$$

Differentiator Amplifier:-



Applying KVL,

$$V' - I_f R_f - V_{out} = 0$$

$$\Rightarrow -I_f R_f = V_{out}$$

$$\Rightarrow V_{out} = -I_f R_f$$

The current equation for capacitor is

$$I_c = \frac{C dV_c}{dt}$$

$\therefore I_c = I_f$

$$\Rightarrow \frac{C dV_{in}}{dt} = \frac{V' - V_{out}}{R_f}$$

$$\Rightarrow C\frac{d}{dt} V_{in} = -\frac{V_{out}}{R_f}$$

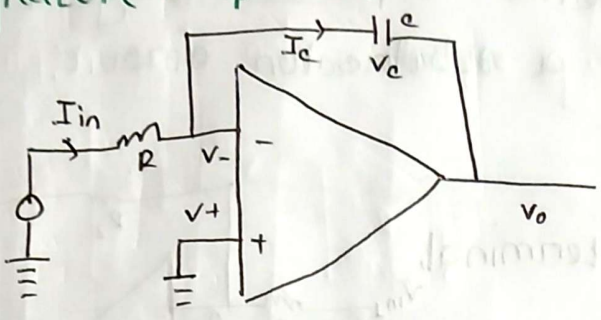
$$\therefore V_{out} = -R_f C \frac{d}{dt} V_{in}$$ (Ans)

Here,

$V_{in} = V_c$

$$\Rightarrow I_c = \frac{C dV_{in}}{dt}$$

Integrator Amplifier:-



Applying KVL,

$$V^- - V_c - V_{out} = 0$$

$$\Rightarrow -V_c = V_{out}$$

$$\Rightarrow V_c = -V_{out}$$

$$\therefore I_{in} = I_c$$

$$\Rightarrow \frac{V_{in} - V}{R} = \frac{C\,dV_c}{dt}$$

$$\Rightarrow \frac{V_{in}}{R} = C\,\frac{d}{dt}(-V_{out})$$

$$\Rightarrow \frac{V_{in}}{R} = -C\frac{dV_{out}}{dt}$$

$$\Rightarrow \frac{d}{dt}V_{out} = -\frac{V_{in}}{RC}$$

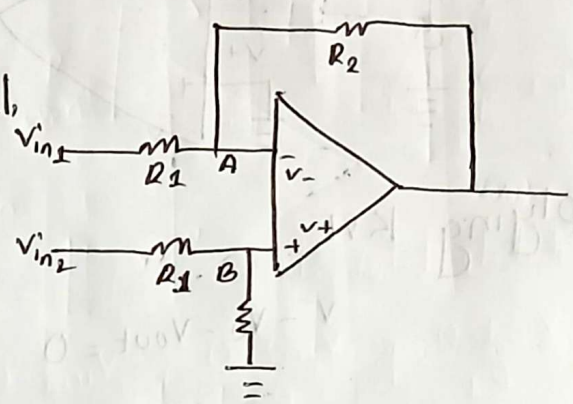
Integrating both side,

$$V_{out}(t) = -\frac{1}{RC}\int V_{in}(t)$$ (Ans)

Subtractor / Differential amplifier using op-amp or op-amp behave as a subtractor circuit:

Step-1

Selecting inverting terminal,



$$\frac{V_{in1} - V_A}{R_1} = \frac{V_A - V_{out1}}{R_2}$$

$$\Rightarrow \frac{V_{in1} - 0}{R_1} = \frac{0 - V_{out1}}{R_2}$$

$$\Rightarrow \frac{V_{in1}}{R_1} = \frac{-V_{out1}}{R_2}$$

$$\therefore V_{out1} = -\left(\frac{R_2}{R_1}\right) V_{in1} \quad \text{---(i)}$$

$V^- - V^+ = 0$

$V^- = V^+$

Also, $V^+ = V_A$

Step-2

Selecting non inverting terminal

Applying voltage divider rule in point B

$$V_B = \frac{R_2 V_{in2}}{R_1 + R_2}$$ [voltage divider rule (VDR) $V_x = \frac{R_x E}{RT}$]

Now by using non-inverting output equation,

$$V_{out2} = \left(1 + \frac{R_2}{R_1}\right) V_B$$

$$= \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_2 V_{in2}}{R_1 + R_2}\right)$$

$$= \frac{R_1 + R_2}{R_1} \times \frac{R_2 V_{in2}}{R_1 + R_2}$$

CamScanner

$\Rightarrow V_{out2} = \frac{R_2}{R_1} V_{in2}$ ——(ii)

(i) + (ii),

$$V_{out1} + V_{out2} = -\frac{R_2}{R_1} V_{in1} + \frac{R_2}{R_1} V_{in2}$$

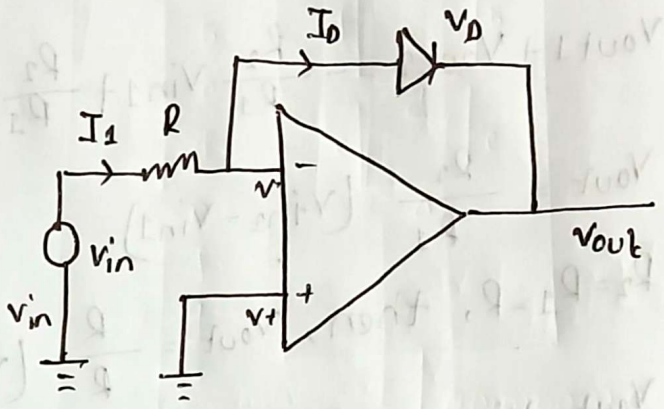
$\Rightarrow V_{out} = \frac{R_2}{R_1} (V_{in2} - V_{in1})$

If $R_2 = R_1 = R$, then, $V_{out} = \frac{R}{R} (V_{in2} - V_{in1})$

$\therefore V_{out} = V_{in2} - V_{in1}$ (Ans)

Log Amplifier:

1. Derive output voltage equation for log amplifier using diode and op-amp:



$I_1 = I_D$

$$\Rightarrow \frac{V_{in} - V^-}{R} = \frac{V_D}{I_0 e^{nV_T}}$$

$$\Rightarrow \frac{V_{in} - 0}{R} = \frac{-V_{out}}{I_0 e^{nV_T}}$$

$$\Rightarrow e^{\frac{-V_{out}}{nV_T}} = \frac{V_{in}}{RI_0}$$

$$\Rightarrow \ln e^{\frac{-V_{out}}{nV_T}} = \ln \frac{V_{in}}{RI_0}$$

$$\Rightarrow V_{out} = nV_T \ln \frac{V_{in}}{RI_0}$$ (Ans)

From diode,
current equation, $I_d = I_0 \left(\frac{V_D}{e^{nV_T}} \right.$

$$= I_0 \frac{V_D}{e^{nV_T}}$$

Here,

n = Ideality factor

V_T = Threshold voltage

V_D = Diode voltage

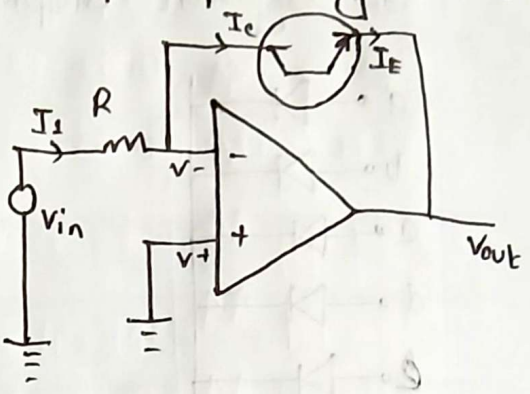
I_0 or I_S = Saturation current

I_d = diode current.

$V^- - V_D - V_{out} = 0$

$\Rightarrow V_D = -V_{out}$.

2. Derive output voltage equation for op-amp using transistor and op-amp:



$I_1 = I_c$

$\Rightarrow \frac{V_{in} - V^-}{R} = I_c$

The collector current equation,

$$I_c = I_s \left(e^{\frac{V_{BE}}{nV_T}} - 1\right)$$

$$= I_s e^{\frac{V_{BE}}{nV_T}}$$

$$\Rightarrow \frac{V_{in}}{R} = I_s e^{\frac{V_{BE}}{nV_T}}$$

Here,

$V_{BE} = V_B - V_E$

$I_E = I_B + I_c$

$I_B \approx 0$

$\therefore I_E = I_c$

Now,

$V_{BE} = V_B - V_E$

$\Rightarrow V_{BE} = 0 - V_E$

$\Rightarrow V_E = -V_{BE}$

$\Rightarrow V_{out} = -V_{BE}$

$\Rightarrow V_{BE} = -V_{out}$.

Here,

n = ideality factor

I_s = Saturation current

V_T = threshold voltage

V_{BE} = Base-emitter voltage

I_c = collection current.

We can write,

$$\frac{V_{in}}{R} = I_s e^{\frac{-V_{out}}{nV_T}} \quad [\text{putting } V_{BE} = -V_{out}]$$

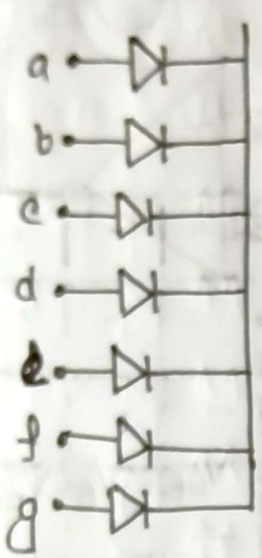
$$\Rightarrow e^{\frac{-V_{out}}{nV_T}} = \frac{V_{in}}{I_s R}$$

$$\Rightarrow \ln e^{\frac{-V_{out}}{nV_T}} = \ln \frac{V_{in}}{RI_s}$$

$$\Rightarrow \frac{-V_{out}}{nV_T} = \ln \frac{V_{in}}{RI_s}$$

$$\Rightarrow -V_{out} = nV_T \ln \frac{V_{in}}{RI_s}$$ (Ans)

7 Segment display for cathode type; BCD



common negative
Ground

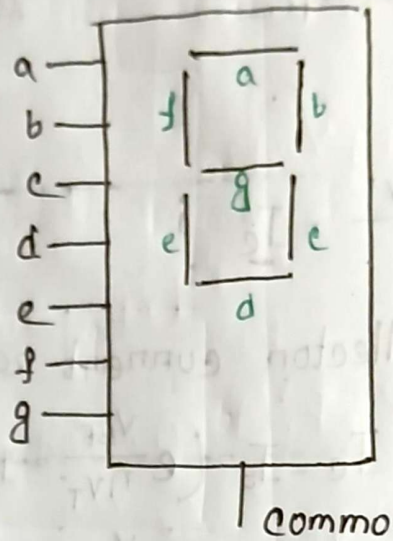
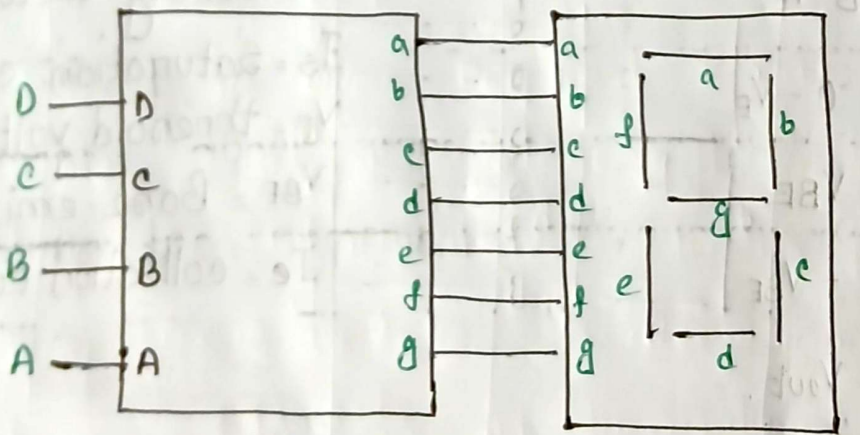
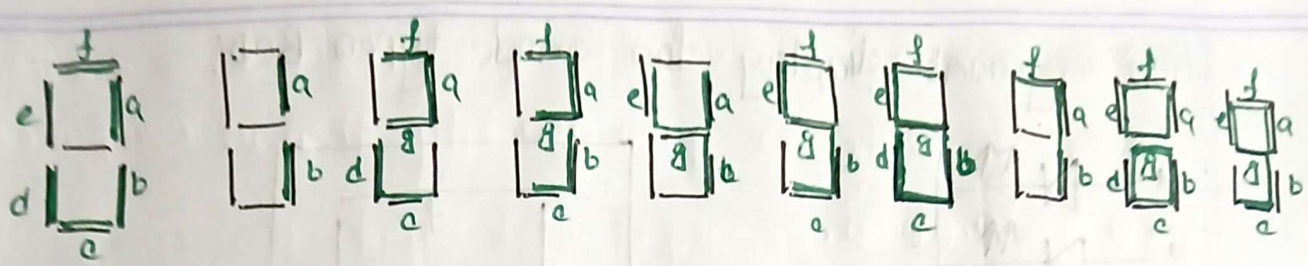


Fig: common cathode display



BCD to 7 segment
Decoder

7 Segment display



Numben	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	1	1	0	0	0	0	0
2	0	0	1	0	1	0	1	1	0	1	1
3	0	0	1	1	1	1	1	0	0	1	1
4	0	1	0	0	1	1	0	0	1	0	1
5	0	1	0	1	0	1	1	0	1	1	1
6	0	1	1	0	0	1	1	1	1	1	1
7	0	1	1	1	1	1	0	0	0	1	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	0	1	1	1

7 segment display for anode type BCD:

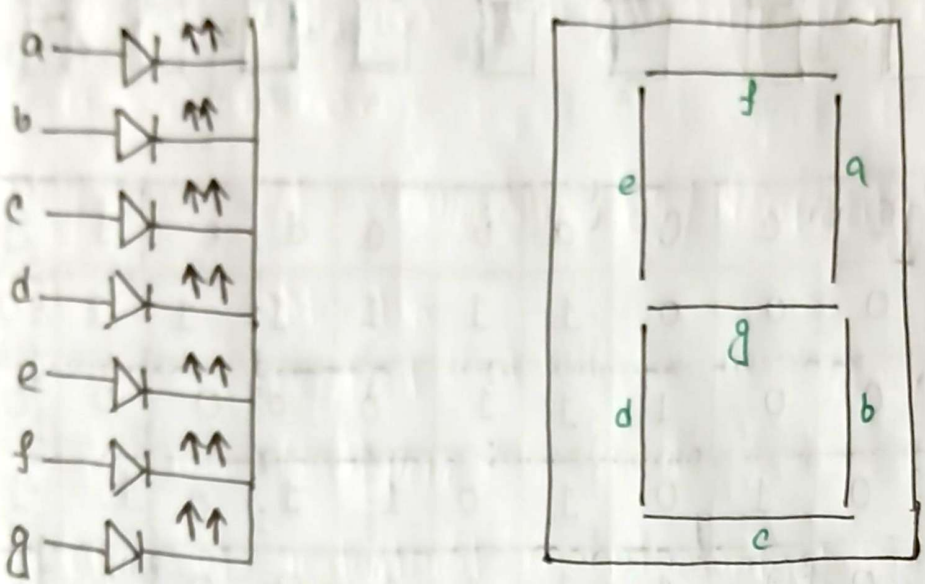
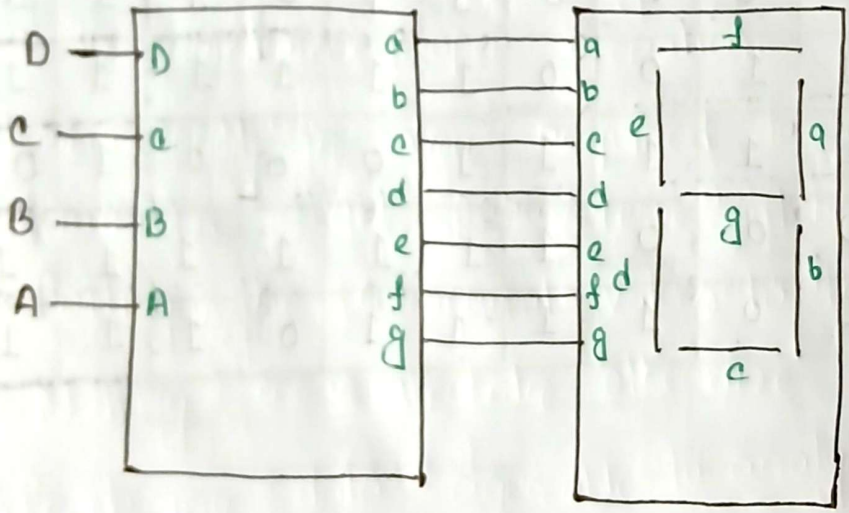
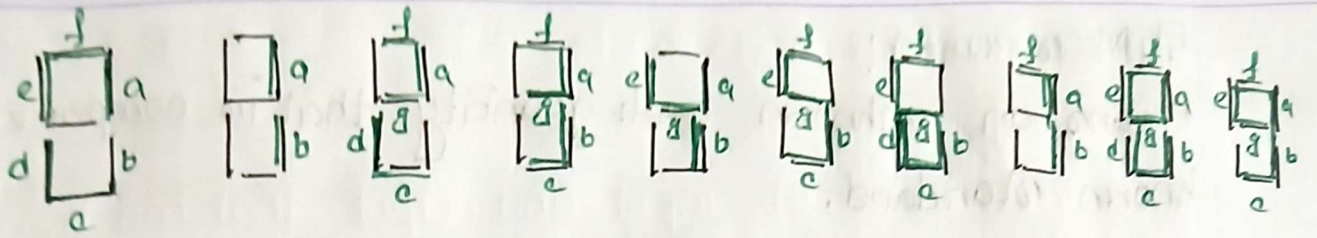


Fig- common anode display



BCD To 7 segment decoden

7 segment display



Number	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	0	0	0	0	0	0	1
1	0	0	0	1	0	0	1	1	1	1	1
2	0	0	1	0	0	1	0	0	1	0	0
3	0	0	1	1	0	0	0	1	1	0	0
4	0	1	0	0	0	0	1	1	0	1	0
5	0	1	0	1	1	0	0	1	0	0	0
6	0	1	1	0	1	0	0	0	0	0	0
7	0	1	1	1	0	0	1	1	1	0	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1							

CMRR:

means common mode rejection ratio. It is given as th ratio of differential mode gain to the common mode gain

$$CMRR = \frac{Ad}{Ac} \qquad CMRR = 20\log_{10}\frac{Ad}{Ac}$$

Voltage Gain:

is a measure of how much electronic amplifier increases the voltage of a signal it receives as input

$$Ac = \frac{V_o}{V_c}$$

V_c = common input signal
V_o = output signal.

Segment-7
Measurement

Aneela Ahmed (C133491)

Measurement:

means an unknown ~~qual~~ quantity that is compared with known standard.

Why we need measurement? / Necessity of measurement:-

→ Monitor
→ Control
→ Analysis

In case of industrial manufacturing:

→ To improve the quality of the product.
→ To improve the efficiency of the product.
→ To maintain proper operation.

Measurement is also required to acquire data on information about parameters interms of

a. Putting the numerical values to the physical quantities
b. Making measurements otherwise inaccessible
c. Producing data agreeable to analysis.

Instrumentation:-

Process of acquiring data about one or more physical quantities using electrical sensors and instruments.

Instrument:-

A device or mechanism used to determine the present value of a quantity under observation.

Accuracy:

The degree of exactness of a measurement compared to the expected value.

Resolution:-

The smallest change in a measured variable to which instrument will response. Also known as "Threshold".

Precision:-

A measure of consistency or repeatability of measurement i.e successive reading do not differ or the consistency of the instrument output for a given value of input. A very precise reading though is not perfectly an accurate reading.

$$\text{Precision} = 1 - \left| \frac{x_n - \bar{x}}{\bar{x}} \right|$$ with x_n = measure value

$\bar{x}$ = average value or expected value.

Expected value:

The design value that is most probable value that calculation indicate one should expect to measure.

Hysterisis:-

The different loading and unloading curve due to the magnetic hysteresis of the iron.

Dead zone:

The total range of possible values for instrument will not given a reading even there is changes in measured

parameter.

Nominal Value:

In some value of input and output that had been stated by the manufacturer for user manual.

Bias:

A constant error that occurs to instrument when point not starting from zero scale.

Range:

A ~~min~~ minimum and maximum range for instrument to operate and it is stated by the manufacturer of the instrument.

Sensitivity:

The ratio of the change in output of the instrument to a change of input or measured variable

$$S = \frac{\Delta \text{output}}{\Delta \text{input}}$$

Steps of generalized measuring system:-

Input → Sensing element → Signal conditioning element → Data Presenting element → Output

Stage: 1

A detection stage (transducer or sensor)

Ex - thermocouple.

Stage: 2

A signal ~~con~~ conditioning stage

Ex: Amplifier, filters, bridges.

Stage: 3

A signal processing stage.

Ex: computer, ADC.

Stage: 4 Read out stage

Ex: printer, oscilloscope.

Methods of measurements:

1. Direct method:-

Easy to do but less accurate

Ex: to measure a steel bar

2. Indirect Method:-

Calibrated system consists of several devices to conver process and display the output.

Ex: To measure from strain gauges located in a structure.

Calibration:-

It implies that there is a numeric relationship throught the whole instrumentation system and that is directly related to an approved national or international standard.

Different types of measuring instruments:

1. Null Type Instruments:-

An instrument in which zero or null indication determines the magnitude of measured quantity such type of instrument is called null type instrument.

Ex: DC potentiometer.

2. Deflection Type instrument:
Deflection provides the basis for measuring the electrical quantity which is known as the deflection type instrument.
Ex: Permanent Magnet moving coil (PMMC) Ammeter.

3. Absolute Instrument:
It gives the values of measured quantities regarding physical constant.
Physical constant means the angle of deflection, degree and meter constant.
Ex- Tangent Galvanometer, Rayleigh's current balance.

4. Secondary instrument:-
The deflection gives the magnitude of electrical quantity to be measured directly
Ex- voltmeter, Thermometer.

5. Active instrument:-
The quantity being measured simply modulates the magnitude of some external power sources.

6. Passive Instrument:-
The instrument output is entirely produced by the quantity being measure.
Ex- Pressure measuring device.

7. Analog instrument:
An analog instrument gives an output that varies continuously as the quantity being measured. Here output is analog form

Ex- Temperature measurement using thermocouple.

8. Digital Instrument:-

A digital instrument has an output that varies in discrete steps and have a finite number of values. Here, output is in digital formm.

Ex - Revalution counter.

Factors consider in selection of instruments:

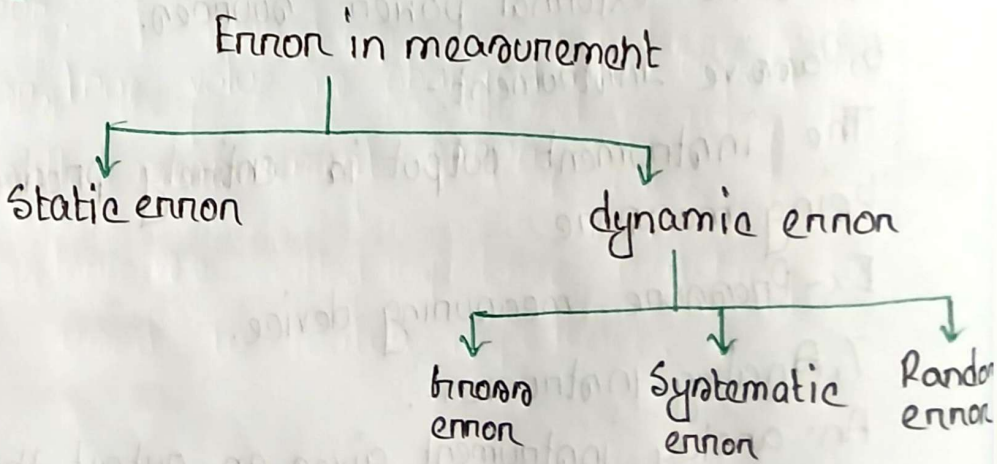
The essential requirements of a measuring instruments are:

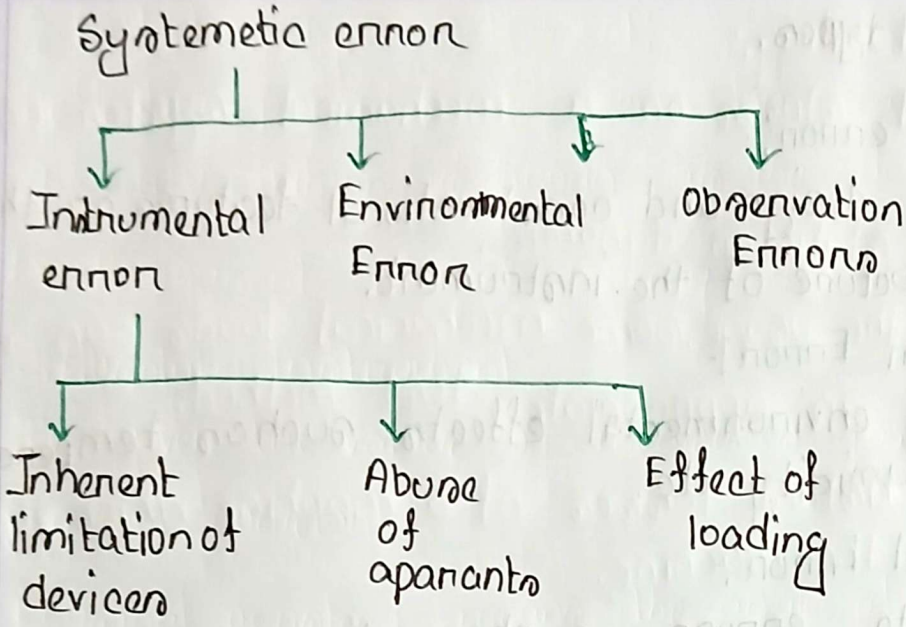
1. Its introduction into the circuit, where measurements are to be made, does not alter the circuit condition.
2. The power consumption for their operation is small.

Error:

is defined as the difference between measured values and true bactual value.

Types of error:-





Static Ennon:-

in the difference betreen meagune value and actual value. Here for a panticulan neading true valuen ane not changed with nespect to time, meann true values ane fixed but meagured valuen ane changed.

$$E_s = V_m - V_t$$

E_s = static ennon
V_m = meagured value
V_t = true value.

It can be both +ve, -ve.

There ane three typen of static ennon, 1. Random ennon
2. Systemic ennon
3. Gnogn ennon.

1. Random ennon:-

These ennons occung due to unknown causen & neasons which ane very difficult to determine

2. Systemic Ennon:-

occuns due to the lack of quality in measuring instruments such as defective equipment on mean & tean effects.

It has three types.

i. Instrumental error:-
occurs due to design and constructional features or the mechanical structure of the instrument.

ii. Environmental Error:-
occurs due to environmental effects such as temperature, pressure, humidity etc.

iii. Observational Error:-
Occurs due to errors in observation.

3. Gross error:-
occurs due to human mistake and error in using measuring instruments or taking wrong readings or mistake in recording observation.

This can't be completely eliminated. But we can try to minimize them by taking proper care while taking reading or recording observation.

Dynamic Error:-
In this error, true value is changed with repeat to time.
It is also called measurement error.

Accuracy vs Precision:-

Accuracy	Precision
1. Closeness of measurement to the true value. Ex: True value = 25mm measured value = 25.01 mm	1. Repeatability of measurement True value = 25mm measured value = 25.5 mm = 25.49 mm = 25.51 mm
2. It have to be precise in most cases.	2. It may on may not be accurate.
3. Affected with systematic error.	3. Affected with random error.
4. Determined by single measurement.	4. To determine precision, several measurements are required.
5. Accuracy can be improved.	5. It can't be improved.

Source of errors:-

1. Lack of knowledge about measurement process.
2. Operational Error.
3. Poor design & contruction.
4. Lack of proper maintainance.
5. Change in process parameters & environmental condition.

Mechanical loading:-

in the physical stress on a mechanical system on component.

loads can be static on dynamic.

Depending on the usage, some mechanical loads can be measured by an appropriate test method.

They called constant losses of a machine which varies with speed of the machine.

Impedence matching:-

It makes sure that the input impedence of one stage called the source, is equal to the output impedance of the following stage called the load.

It helps to

→ Transfer maximum power to the network on load.

→ Improve the signal to noice ratio of the system.

→ Matching network is ideally lossless.

The advantage of transformer as impedence matching device is that they have broadband meaning they can work with a wide range of frequencies.

Special transformers called impedence matching transformers can be used to match impedence.